

Report

Analysis on Netflix User base

I. Introduction –

Netflix, one of the world's leading streaming platforms, has revolutionized the way people consume entertainment. With a subscriber base spanning over 190 countries, Netflix offers a vast library of movies, TV shows, and original content tailored to diverse audiences. Its success largely stems from its data-driven approach to understanding user behaviour, enabling personalized recommendations and strategic content decisions.

The report is structured as follows: first, an overview of the dataset and methodology; second, a presentation of the analysis results; third, a discussion of the findings; and finally, recommendations and conclusions based on the insights derived from the data.

II. Methodology –

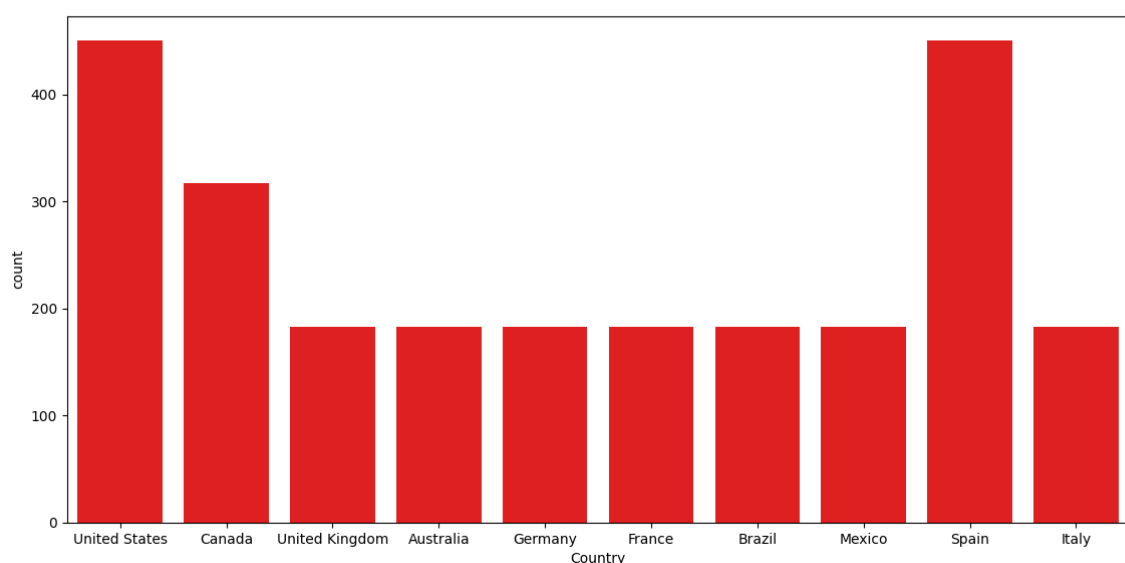
Data Source – This is primary Data set, this data set has a information about Subscription Types, Revenue, User Details, and Activity. The Data set is provided by Netflix User Base Dataset on Kaggle website.

Tools – Statistical Analysis, Case Study.

III. Adding visuals –

A Cluster Column showing number of subscription in a country.

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
data=pd.read_csv("C:/Users/IES/Downloads/Netflix Userbase.csv")
sns.countplot(data=data, color='red',x="Country")
plt.show()
```



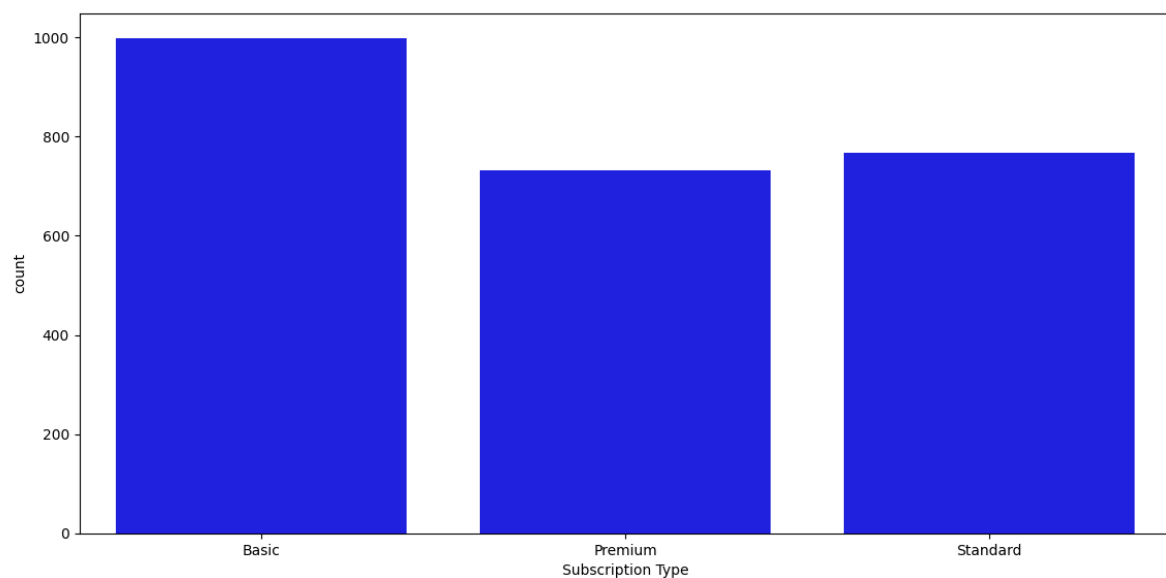
Why USA and Spain has a highest number Netflix user?

USA: The USA has a large population with high internet penetration, making it an ideal market for streaming services. Widespread availability of high-speed internet and affordable devices such as smartphones, TVs that supports streaming services like Netflix. Netflix offers a vast library of English-language content, catering to the American audience's preferences. Original productions like Stranger Things and House of Cards resonate strongly with U.S. viewers. A significant portion of the U.S. population can afford subscriptions, making the market lucrative for Netflix.

SPAIN: Spain has experienced rapid growth in digital infrastructure, with widespread access to affordable internet services. Netflix invests in producing Spanish-language content, such as La Casa de Papel (Money Heist) and Elite, which are immensely popular in Spain and globally. This local focus drives higher user engagement and subscriptions. Changing consumer habits, particularly among younger audiences, have made streaming platforms a preferred choice over traditional TV in Spain. Spanish users enjoy a mix of global hits and culturally relevant shows, which Netflix provides.

A Cluster Column showing type of subscription is used by User.

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
data=pd.read_csv("C:/Users/IES/Downloads/Netflix Userbase.csv")
sns.countplot(data=data, color='blue',x="Subscription Type")
#plt.lineplot()
plt.show()
```

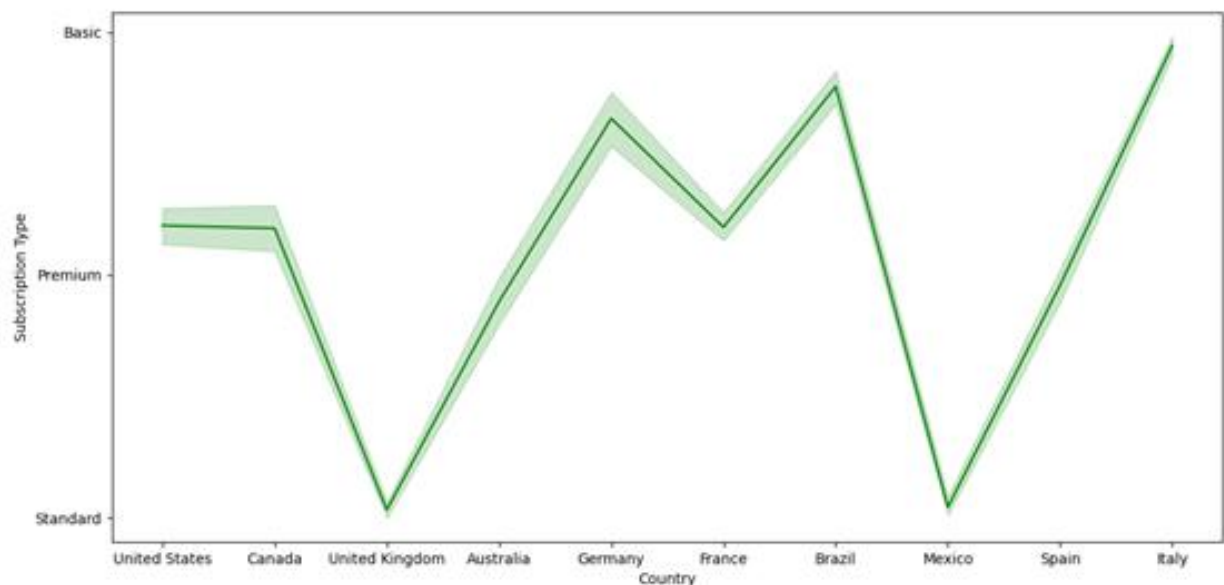


Why most of the user use basic version of in Netflix?

The Basic plan is the most affordable option, making it attractive to cost-conscious users. In regions with lower average incomes, the Basic plan is often the most feasible choice. Some users primarily use devices like older smartphones or low-resolution laptops where HD or UHD quality doesn't make a significant difference. New subscribers might choose the Basic plan as an entry-level option to explore Netflix before considering an upgrade.

A Line Plot showing which country has highest number of Basic, Standard and premium type of Subscription.

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
data=pd.read_csv("C:/Users/IES/Downloads/Netflix Userbase.csv")
sns.lineplot(data=data, color='green',x="Country",y="Subscription Type")
plt.show()
```



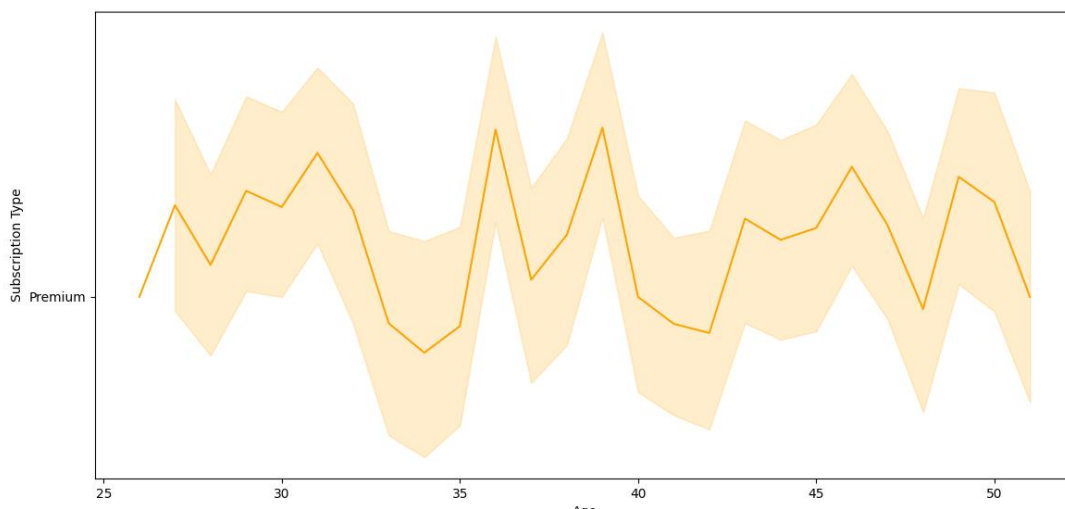
Why most of people in Italy use basic type of subscription in Netflix?

Competition and Alternatives: In Italy, Netflix faces strong competition from other local streaming services like Disney+, Amazon Prime Video, and RaiPlay. These alternatives sometimes offer cheaper plans, bundled services, or content more tailored to Italian audiences, which can make users more cautious when choosing to spend more on a premium Netflix plan.

Cost Sensitivity: The basic plan is the most affordable option, with a lower monthly fee compared to the standard and premium plans. In Italy, like many other countries, cost is a significant factor for consumers when deciding on subscriptions, and many people prefer to opt for the cheapest option to save money.

A Line Plot showing most of the people between 35 to 40 use premium subscription in Netflix.

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
data=pd.read_csv("C:/Users/IES/Downloads/Netflix Userbase.csv")
sns.lineplot(data=data, color='orange',x="Age",y="Subscription Type")
plt.show()
```



Why most of the people between 35 to 40 use premium subscription in Netflix?

Family and Household Use -

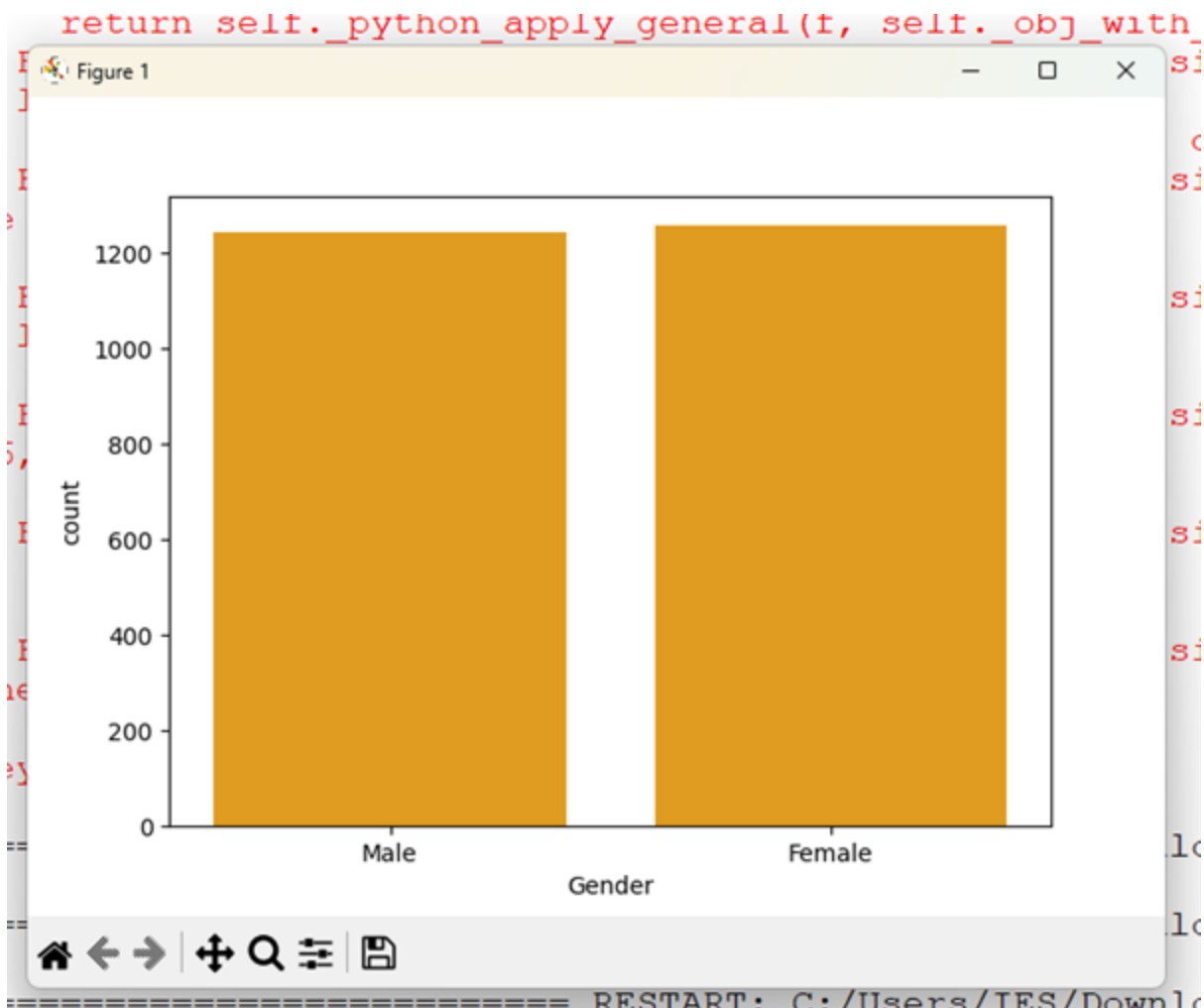
- **Multiple Screens:** People in the 35-40 age range are often at a stage in life where they have families or are living in households with multiple people (partners, children, roommates, etc.). The premium plan allows streaming on up to four devices simultaneously, which is ideal for families or households with more than one person who wants to watch content at the same time.
- **High Demand for Content:** With children, partners, or roommates involved, there may be more demand for simultaneous streams across different devices (smart TVs, tablets, laptops, etc.), making the premium plan a more practical option.

Lifestyle and Viewing Habits -

Streaming Preferences: People in this age group are more likely to consume content on various devices, including multiple TVs or streaming on mobile devices. The premium plan's flexibility in supporting several devices for simultaneous streaming fits their dynamic viewing habits.

A Cluster Bar showing user of Netflix by gender

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
data=pd.read_csv("C:/Users/IES/Downloads/Netflix Userbase.csv")
sns.countplot(data=data, color='orange',x="Gender")
#plt.lineplot()
plt.show()
```



Why female use Netflix more than male?

Female-led stories and strong female protagonists are common in Netflix Originals, attracting a larger female audience. Studies show that women are often more interested in true crime and real-life documentaries, a genre heavily featured on Netflix. Female audiences often engage more in discussions around popular Netflix series on social media, leading to increased usage to stay up-to-date. Shows like *Gilmore Girls* and *Sex Education* have strong cult followings among women. In many households, women play a key role in selecting streaming subscriptions and choosing what to watch, influencing overall usage statistics.

IV. Conclusion –

The analysis of Netflix user data provides valuable insights into user behaviour, preferences, and engagement patterns. Key findings reveal that demographics such as Country, gender, and Subscription type significantly influence content preferences and subscription choices. For instance, female users show higher engagement, favouring genres like drama and romance, while younger users demonstrate a strong preference for binge-watching trending series.

Viewing habits indicate that most users prefer the Basic plan due to its affordability and compatibility with individual streaming needs. Additionally, regional differences, such as high user concentrations in the USA and Spain, highlight the importance of localized content strategies.

V. Recommendation –

1. Expand Localized Content

- Invest in region-specific content to cater to diverse user bases, as evidenced by the popularity of shows like *La Casa de Papel* in Spain and other localized productions.
- Increase collaborations with local creators to produce culturally relevant content.

2. Enhance Personalization

- Improve recommendation algorithms to account for subtle user preferences, such as favourite subgenres or niche interests.
- Introduce features that allow users to customize their viewing profiles further.
Example - preferred genres, content suggestions based on mood.

3. Monitor and Address Churn

- Use predictive analytics to identify users at risk of cancelling subscriptions and offer incentives. Example - temporary discounts, exclusive previews to retain them.
- Conduct surveys to understand reasons for churn and address recurring issues.

4. Invest in Emerging Markets

- Focus on expanding in regions with growing internet penetration and untapped potential, such as parts of Asia, Africa, and Latin America.

5. Improve Accessibility

- Provide multi-language subtitles and dubbing for popular content to cater to non-English speaking audiences.